

DAVID IHOMBA KIBANDI

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with a foundation in backend development and Information Science, specializing in building scalable, secure applications with Python, Flask, and SQLAlchemy, and responsive interfaces with React.js. Recently designed and shipped Deliveroo, a full-stack parcel delivery platform featuring JWT authentication with role-based access control, RESTful APIs, real-time courier tracking with Google Maps integration, and automated email notifications, now live in production. Brings a distinctive combination of technical rigor and a background in media production and public relations, with strong attention to detail, workflow optimization, and clear communication with cross-functional teams.

TECHNICAL SKILLS

Languages & Frameworks: Python, Flask, JavaScript, React.js, Redux Toolkit / React Context, Web Development, Software Development

Databases & ORM: PostgreSQL, SQLAlchemy, Marshmallow, Database Security & Design

APIs & Integrations: RESTful API Design, JWT-Based Authentication, Secure Authentication, Role-Based Access Control (RBAC), WebSocket, Google Maps API, Resend (Email API), Electronic Payment Processing (Mock M-Pesa Integration)

Engineering Practices & Tools: Git/GitHub, Figma (Wireframing), Agile/Team-Based Development, ERD & Relational Database Design, Software Engineering Practices, Project Management, Cross-Functional Team Leadership

Information & Media Management: Knowledge & Information Management, Information Management Solutions, Records Management, Library Systems & Services, Digital Libraries, Media Management, ICT & Media Technician, ICT Facilitation, Video Production

Communication: Strategic Communications, Interpersonal Communication, Public Relations & Stakeholder Communication

FEATURED PROJECT

Deliveroo — Full-Stack Parcel Delivery Platform | *Independent / Capstone Group Project* 2026

- Designed the relational database schema (ERD) supporting three user roles Customer, Courier, and Admin with tracked courier reassignment history and a customer-to-courier rating system.
- Implemented JWT-based authentication with role-based access control and hashed passwords to secure all API endpoints.
- Built RESTful APIs in Flask for order lifecycle management (Pending, Picked Up, In Transit, Delivered, Cancelled), including cancellation logic restricted to pre-pickup orders.
- Integrated Google Maps for real-time courier location tracking, route polylines, and distance/ETA calculation, using WebSocket connections for live updates, plus automated email notifications via Resend for status changes and delivery updates.
- Developed pricing logic and mock electronic payment processing (M-Pesa) based on parcel weight category and delivery distance, with itemized cost breakdown, and an admin dashboard for delivery statistics and courier performance.
- Built the frontend in React with Redux Toolkit for state management, following mobile-friendly Figma wireframes, while coordinating cross-functionally with a group development team.

Live: <https://deliveroo-capstone-project.vercel.app/>

PROFESSIONAL EXPERIENCE

Media Technician / ICT Facilitator | *Pillarframe Media House* Apr 2017 – Present

- Managed end-to-end digital media projects from client consultation through final delivery, ensuring technical and creative specifications were consistently met.
- Worked as videographer and photographer, and led video production for client events and campaigns.
- Served as ICT facilitator, supporting equipment setup, technical troubleshooting, and system readiness for media production.
- Processed and refined over 100 video and photo projects using advanced digital editing software, improving final content quality and client satisfaction.

- Designed and implemented streamlined workflows for media asset management and online publishing, increasing project delivery efficiency by 15%.
- Translated client creative concepts into detailed technical requirements, aligning final digital products with brand vision and strategic goals.

News Gathering & Studio Technician | *Kenya Broadcasting Corporation (KBC)* Jan 2016 – Sep 2018

- Operated and maintained studio broadcasting equipment for live prime-time news, achieving 99.9% operational uptime during critical transmissions.
- Produced complete news packages by gathering, editing, and integrating multimedia sources under strict production deadlines.
- Configured and managed studio systems for live programs, optimizing technical workflows for production crews and on-air talent.
- Collaborated with journalists and producers to translate editorial requirements into precise technical execution.

Election Clerk | *Independent Electoral and Boundaries Commission* Aug 2017

- Operated electronic voter identification systems, accurately processing over 500 voters while maintaining 100% data integrity.
- Executed data handling and verification tasks under strict procedural protocols to support a secure electoral process.
- Performed system diagnostics and pre-operational checks on assigned electronic equipment to prevent workflow disruptions.

EDUCATION

Certificate in Software Engineering (Full-Stack Development) | *Moringa School* 2026

B.Sc. Information Science | *Karatina University* Aug 2013 – May 2019

Diploma in Journalism and Mass Communication | *Mount Kenya University* Jan 2015 – Nov 2016